Geometry of Multi-Turn Conversation Trajectories in Small Language Models: Characterization, Compression, and Limits

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Abstract*—We present a manuscript-grade checkpoint of a three-stage research program on conversational memory compression for small language models and, equally importantly, a record of where that program fails. The program proceeds from geometry measurement (Paper 1), to geometry-aware control under fixed memory budgets (Paper 2), to sparse conversational memory codecs (Paper 3). In Paper 1, we model turn-level conversation states as points on the unit sphere and analyze transported increments in local tangent spaces. Across three compact model families, conversation-state trajectories are extremely low-rank and geometric reconstruction error is tightly coupled to decoder drift: mean rank95 ranges from 1.049 to 1.528, while the correlation between geometric distortion and logit drift ranges from 0.989 to 0.994, with permutation and shuffled-order null controls collapsing these effects. In Paper 2, geometry-aware budget allocation improves decoder fidelity relative to uniform retention under conversational memory scarcity and preserves memory-critical support turns more often than uniform policies on a hard stress set. In Paper 3, sparse codecs become real on the same hard support-turn benchmark family: fairness-controlled `geometry_keep_compress_drop` is strongest under low-to-mid budgets on the hard set, while plain geometry retakes the lead in a looser-budget 3B regime. But the broader benchmark story is not a universal geometry win. On the full LongMemEval public benchmark, semantic or segment-style policies are strongest under scarcity and KCD becomes best only at 0.50 budget; on MSC-style conversational-memory benchmarks, plain semantic retention remains the strongest family. We also report explicit negative results: geometry does not robustly separate support/persona turns from filler in MSC, a geometry-only regime atlas remains too coarse to classify conversation types reliably, and two state-update supersession formulations fail on a synthetic falsification benchmark. The resulting claim boundary is sharper and stronger: geometry is real, decoder-relevant, and useful for support-turn-critical memory control, but semantic-memory benchmarks require semantic-led policies, and several plausible geometry-only extensions fail in compact models.

***Index Terms*—large language models, long-context memory, conversational memory, representation geometry, memory compression, sparse codecs**

I. Introduction

Long multi-turn conversations create a memory problem before they create a generation problem. Once the full dialogue no longer fits inside a practical inference budget, the model must decide what to retain, what to compress, and what to evict. The central claim of this repository checkpoint is that *hidden-state geometry of the evolving conversation trajectory* is a useful control signal for that decision.

This repository organizes the project into three linked papers:

1) **Paper 1: Geometry characterization.** Determine whether turn-level conversation-state trajectories are low-rank, curved in informative ways, and predictively tied to decoder behavior.
2) **Paper 2: Geometry-aware control.** Use geometry-derived risk scores to allocate a fixed memory budget.
3) **Paper 3: Sparse conversational codecs.** Replace part of raw history retention with a compressed memory object that preserves the right support structure.

The current checkpoint is beyond proof-of-concept, but it is not a universal-win story. Paper 1 is frozen scientifically; Paper 2 has a stable systems result with a mechanism story; Paper 3 has real sparse codecs on hard support-turn benchmarks, but also clear benchmark-dependent losses and several failed extensions. The main open problem is no longer whether geometry matters. It is how to combine geometry with semantic signal in a way that respects the type of memory the benchmark actually requires.

a) Main contributions of the checkpoint.:

- A geometric formalization of turn-level conversational dynamics on the sphere using local tangent-space transport.
- Stable empirical evidence that conversation-state trajectories are extremely low-rank and strongly decoder-relevant across compact model families.
- A geometry-aware memory-control result under conversational scarcity, including a mechanism analysis showing rescue of memory-critical support turns.
- A first fairness-controlled sparse codec family, `geometry_keep_compress_drop` (KCD), that is active, budget-responsive, and positive on hard support-turn benchmarks.
- An explicit benchmark-dependent Paper 3 reading: semantic-memory benchmarks favor semantic-led policies, while support-turn-critical benchmarks favor geometry-aware codecs.
- A falsification record showing the limits of geometry-only extensions in compact models: failed persona-vs-filler

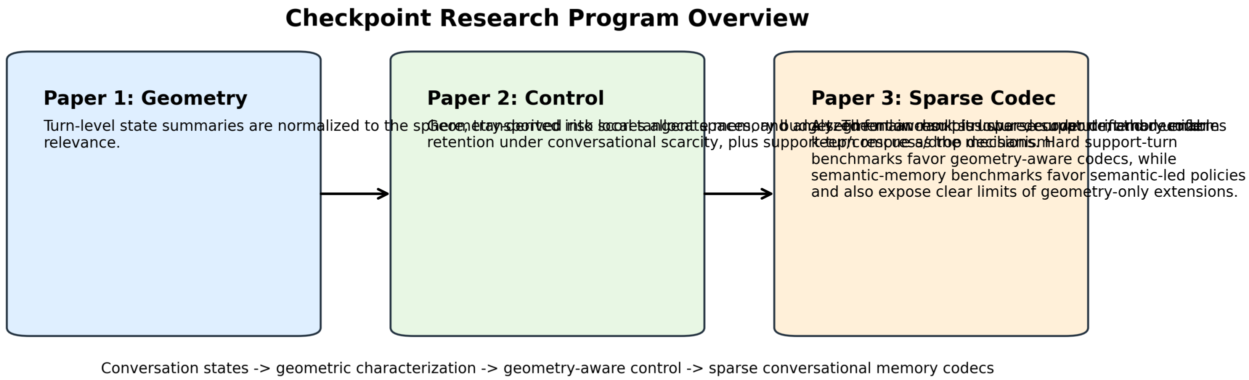

Fig. 1. Research program overview. Paper 1 establishes low-rank, decoder-relevant conversation-state geometry. Paper 2 converts geometry into a budget allocation signal. Paper 3 turns that signal into a sparse keep/compress/drop memory codec.

separation on MSC, a too-coarse geometry-only regime atlas, and failed state-update supersession detection.

II. Mathematical Setup

A. Conversation-State Geometry

For a multi-turn conversation $x_{1:T}$, let

$$z_t = f_\theta(x_{1:t}) \in \mathbb{R}^d \tag{1}$$

be a turn-level hidden-state summary extracted from the model after turn t. We normalize onto the sphere:

$$h_t = \frac{z_t}{\|z_t\|} \in \mathbb{S}^{d-1}. \tag{2}$$

Within a short segment S_j, choose a reference point $q_j \in \mathbb{S}^{d-1}$. The one-step spherical increment from h_t to h_{t+1} is mapped into the tangent space at q_j:

$$u_t = \mathrm{PT}_{h_t \to q_j}\big(\log_{h_t}(h_{t+1})\big) \in T_{q_j}\mathbb{S}^{d-1}, \tag{3}$$

where $\log$ is the Riemannian logarithm map and PT is parallel transport. Paper 1 tests whether these transported increments are approximately low-rank:

$$u_t \approx B_j c_t + e_t, \qquad B_j \in \mathbb{R}^{d \times r_j}, \quad r_j \ll d. \tag{4}$$

We use the following discrete curvature proxy:

$$\kappa_t \approx \frac{\|u_t - u_{t-1}\|_2}{\Delta s_t}, \tag{5}$$

with Δs_t the local geodesic step length. A large κ_t indicates locally unstable geometry or a potential conversational regime transition.

B. Decoder-Relevance Metrics

Given a reconstructed state trajectory $\widehat{h}_{1:T}$, decoder distortion is measured by comparing the full and compressed decoder outputs:

$$\Delta_{\ell_2} = \|\ell(\widehat{h}_t) - \ell(h_t)\|_2, \tag{6}$$

$$\mathrm{KL} = \mathrm{KL}\Big(p_\theta(\cdot \mid h_t) \,\|\, p_\theta(\cdot \mid \widehat{h}_t)\Big), \tag{7}$$

$$\mathrm{Top1} = \mathbf{1}\left[\arg\max p_\theta(\cdot \mid h_t) = \arg\max p_\theta(\cdot \mid \widehat{h}_t)\right]. \tag{8}$$

Paper 2 and Paper 3 additionally evaluate answer-level average negative log-likelihood relative to the gold next assistant response.

C. Budgeted Memory Control

For a memory policy π acting on conversation units u, the control problem is

$$\min_\pi \mathbb{E}[D(\pi)] \quad \text{s.t.} \quad \mathbb{E}[M(\pi)] \le B, \tag{9}$$

where $D(\pi)$ is decoder distortion and $M(\pi)$ is realized memory cost. In Lagrangian form:

$$\mathcal{L}(\pi) = \mathbb{E}[D(\pi)] + \lambda\, \mathbb{E}[M(\pi)]. \tag{10}$$

Paper 2 instantiates π as a turn- or segment-level retention rule driven by a geometry score. Paper 3 turns π into a sparse memory-action policy

$$a_j \in \{\mathrm{keep}, \mathrm{compress}, \mathrm{evict}\} \tag{11}$$

over segment j, producing a compressed memory object

$$m_j = (a_j, s_j, \mathrm{meta}_j), \tag{12}$$

where a_j is the anchor/keyframe summary, s_j is sparse support memory, and meta_j stores role/order information.

III. Theory Directions and Proof Sketches

A. Paper 1: Transported Low-Rank Geometry

Proposition 1. *Let $h_t \in \mathbb{S}^{d-1}$ be a conversation-state trajectory and let u_t be its transported increments in a common tangent space $T_{q_j}\mathbb{S}^{d-1}$. Then local low-rank analysis of the sequence $\{u_t\}_{t \in S_j}$ is well-defined independently of the original ambient coordinates, up to the choice of segment reference point and the local transport gauge.*

Proof sketch. Parallel transport is an isometry between tangent spaces on the sphere. Therefore norms and principal angles between transported increments are preserved under the transport map. Once all increments in a segment are moved into the same tangent space, ordinary Euclidean low-rank analysis becomes meaningful without mixing incompatible local coordinates. □

Theorem 1. *Assume bounded segment curvature $\kappa_t \leq \kappa_{\max}$ and subspace approximation error $\|e_t\| \leq \varepsilon$ in each segment. Then piecewise low-rank reconstruction error on the sphere is bounded by a second-order curvature term plus the accumulated subspace error:*

$$d_{\text{geo}}(h_t, \widehat{h}_t) \leq C_1 \kappa_{\max} L_j^2 + C_2 \sum_{t \in S_j} \varepsilon_t, \tag{13}$$

where L_j is segment length in geodesic arc length.

Proof sketch. Approximate the trajectory within each segment by integrating the projected tangent increments. Geodesic integration on the sphere has second-order local truncation error in curvature. The projection residual contributes linearly through Grönwall-style accumulation along the segment. Summing the two terms yields the stated bound. □

B. Paper 2: Priority Allocation Under Scarcity

Proposition 2. *Suppose decoder harm is monotone in a geometry-derived risk score $r(u)$ and memory cost is additive across units. Then for fixed budget B, allocating memory to higher-risk units before lower-risk units weakly dominates uniform allocation in expected decoder distortion.*

Proof sketch. Under monotone marginal harm, the cost-constrained optimization becomes a knapsack-style prioritization problem. Any reallocation that moves memory from a lower-risk unit to a higher-risk unit cannot increase expected harm. Iterating this exchange argument yields weak dominance over uniform allocation whenever the risk ordering is informative. □

C. Paper 3: Sparse Codec Stability

Theorem 2. *Assume (i) the decoder is locally Lipschitz in a decoder-aware geometry metric, (ii) compressed segment representations preserve anchors and the latest support turn with bounded reconstruction error, and (iii) omitted residual information is bounded by a fixed exogenous segment-wise harm predictor. Then decoder drift under keep/compress/drop is bounded by the sum of per-segment codec distortion terms.*

Proof sketch. The decoder Lipschitz condition turns state reconstruction error into logit drift. The sparse codec decomposes reconstruction error into an anchor term, a support-memory term, and an evicted-residual term. Bounding each component separately and summing over segments yields a codec-level distortion bound. The theorem is intentionally directional: the current checkpoint validates the assumptions empirically but does not yet close them with a learned distortion model, and the harm predictor in assumption (iii) is still an explicit assumption rather than a learned component. □

TABLE I
CURRENT MODEL REGISTRY USED ACROSS THE CHECKPOINT.

Preset	Hugging Face model	Current role
qwen25_05b	Qwen/Qwen2.5-0.5B-Instruct	Paper 1/Paper 2 local baseline
qwen25_15b	Qwen/Qwen2.5-1.5B-Instruct	Main fairness-controlled model
qwen25_3b	Qwen/Qwen2.5-3B-Instruct	First 3B Paper 2/Paper 3 validation
smollm2_17b	HuggingFaceTB/SmolLM2-1.7B-Instruct	Non-Qwen compact control family
llama32_3b	meta-llama/Llama-3.2-3B-Instruct	Non-Qwen 3B public-benchmark runner

IV. Experimental Design

A. Models

We evaluate compact and mid-scale open models suitable for local and Colab-based study. Table I summarizes the current preset registry.

B. Conversation Families and Evaluation Sets

Paper 1 uses audited short multi-family conversations for geometric characterization and changepoint analysis. **Paper 2** uses a harder long-context stress set with three families: long dependency, retrieval-heavy, and code conversation. The hard set is deliberately constructed so that late-turn answers depend on earlier facts or constraints. **Paper 3** reuses the same hard set and then extends it with fairness-controlled sweeps and 3B validation.

C. Policies

We compare the following policy families.

- **uniform**: budget spread without geometry.
- **geometry**: direct geometry-based retention.
- **semantic**: a conversation-level semantic competitor for Paper 2 and public-benchmark extensions.
- **geometry_segment_actions**: segment keep/compress/evict bridge policy.
- **geometry_keep_compress_drop (KCD)**: sparse codec family preserving anchors and sparse support memory.

D. Statistical Protocol

The repository computes bootstrap standard deviations and 95% confidence intervals for key means, along with paired sign-flip or paired-difference tests relative to uniform and pairwise policy comparisons where applicable. This manuscript reports only tracked results already present in versioned artifacts.

V. Results

A. Paper 1: Conversation-State Geometry Is Compact and Decoder-Relevant

The core Paper 1 result is strong and stable. Conversation-state trajectories are extremely low-rank across all tested models, with mean rank95 between 1.049 and 1.528. Shuffling turn order substantially increases rank95, which rules out the most obvious null explanation that low-rank structure is merely an artifact of bounded dimensionality or summary extraction. At the same time, geometric distortion is almost perfectly aligned with output-space drift.

The exact numbers are the right scale for a compressibility argument. For `qwen25_05b`, mean rank95 is 1.049

TABLE II
PAPER 1 MAIN QUANTITATIVE RESULTS. RANK95 REMAINS SMALL, SHUFFLED-TURN NULLS INCREASE RANK, AND GEOMETRY–LOGIT COUPLING REMAINS EXTREMELY HIGH.

Model	Rank95	95% CI	Shuffled Rank95	p_{H1}	Corr($d_{\mathrm{geo}}, \Delta\ell$)	p_{H3}
qwen25_05b	1.049	[1.000, 1.111]	1.486	0.0000	0.989	0.0000
qwen25_15b	1.167	[1.083, 1.271]	1.458	0.0008	0.994	0.0000
smollm2_17b	1.528	[1.500, 1.576]	1.799	0.0122	0.989	0.0000

TABLE III
PAPER 2 GEOMETRY-VS-UNIFORM RESULTS ON THE HARD STRESS SET. NEGATIVE DELTAS ARE IMPROVEMENTS OVER UNIFORM.

Model	ΔLogit@0.20	p	ΔLogit@0.35	p	ΔNLL@0.35	p
qwen25_05b	-65.654	0.0000	-99.503	0.0000	-0.091	0.7222
qwen25_15b	-33.647	0.0015	-21.669	0.2637	-0.181	0.1343
smollm2_17b	-4.105	0.2878	-11.329	0.0200	-0.002	1.0000

with a 95% bootstrap interval of [1.000, 1.111], while the shuffled-turn-order null rises to 1.486. The corresponding geometry–logit correlation is 0.989. The same pattern holds for `qwen25_15b` (rank95 1.167, correlation 0.994) and `smollm2_17b` (rank95 1.528, correlation 0.989). These are not borderline effects. They define the scientific core of the program.

The boundary-localization hypothesis remains secondary. Geometry carries regime-boundary information, but current evidence supports approximate localization rather than exact segmentation. That result is scientifically useful because it tells later stages to prefer soft risk allocation over brittle exact regime decoding.

B. Paper 2: Geometry-Aware Control Beats Uniform Under Scarcity

Paper 2 asks whether the geometry signal can be used as a control law under fixed memory budgets. On the hard stress set, the answer is yes: geometry-aware control reduces decoder distortion relative to uniform retention under scarcity. The effect is strongest in the low-to-mid budget regime.

At budget 0.20, geometry improves mean logit drift by 65.65 on `qwen25_05b` and 33.65 on `qwen25_15b`; at budget 0.35, the improvements are 99.50 and 21.67, respectively. The `qwen25_15b` effect weakens faster than the `qwen25_05b` effect, but the low-to-mid budget regime still carries the clearest overall Paper 2 signal. `smollm2_17b` is noisier, but still directionally favorable in the same region. The harder behavior probe is no longer flat, although answer-level improvements remain mixed and weaker than logit-level improvements.

The mechanism story is now one of the strongest parts of the repo. On the focused `qwen25_05b` case analysis at budget 0.35, geometry retains more support user turns than uniform in 14/36 cases, while uniform does so in only 5/36. Geometry keeps the latest support turn while uniform drops it in 13/36 cases. The rescued turn types are exactly the ones that matter for downstream reasoning: exact code constraints, SQL ordering requirements, retrieval logistics, and short base-memory fact bundles.

TABLE IV
PAPER 3 FAIRNESS AND 3B CHECKPOINT. KCD IS STRONGEST IN THE SCARCITY REGIME; GEOMETRY RETAKES THE LEAD WHEN BUDGET LOOSENS ON THE 3B PROBE.

Setting	Policy	Budget	ΔLogit	p	ΔNLL	p
qwen25_15b fairness	geometry_keep_compress_drop	0.24	-38.272	0.0022	-0.034	0.5282
qwen25_15b fairness	geometry_keep_compress_drop	0.32	-56.165	0.0030	-0.390	0.0270
qwen25_15b fairness	geometry_keep_compress_drop	0.50	7.050	0.6730	-0.638	0.0040
qwen25_3b probe	geometry_keep_compress_drop	0.35	-70.997	0.0070	–	–
qwen25_3b probe	geometry	0.50	-46.526	0.0333	–	–

The correct claim boundary is therefore narrow and strong: geometry is the right signal for decoder-fidelity control in conversations, but not every downstream answer-level metric has yet become geometry-specific. That is acceptable for the current paper boundary because the primary systems objective is decoder-stability under budget.

C. Paper 3: Sparse Conversational Codecs Are Real, But Not Universal

Paper 3 is where the program crosses from smarter retention into genuine compression. The positive result is real: KCD is no longer degenerate. The `compress` branch is active, budget-responsive, and mechanism-valid on hard support-turn benchmarks. But the broader lesson is no longer "geometry codec wins everywhere." The winning policy depends on the kind of memory the benchmark requires.

On the fairness-controlled hard-set `qwen25_15b` sweep, KCD is the strongest logit policy through the low/mid-budget region, with particularly strong results at:

- 0.24: $\Delta_{\ell_2} = -38.272$, relative logit 0.898, $p = 0.0022$
- 0.28: $\Delta_{\ell_2} = -42.138$, relative logit 0.883, $p = 0.0008$
- 0.32: $\Delta_{\ell_2} = -56.165$, relative logit 0.845, $p = 0.0030$
- 0.38: $\Delta_{\ell_2} = -41.881$, relative logit 0.871, $p = 0.0213$

These are fairness-relevant because realized token fractions are tight in the same region: at budget 0.24, KCD uses slightly *less* realized token fraction than uniform while still beating it on decoder distortion.

Behavior also strengthens in the same regime. KCD produces significant answer-level improvements on `qwen25_15b` at 0.32, 0.35, 0.46, and 0.50. Pairwise results show that KCD beats `geometry_segment_actions` on behavior at 0.46 and 0.50.

The 3B probe sharpens the hard-set picture further. On `qwen25_3b`, KCD is the strongest codec policy at 0.35, but plain geometry becomes the strongest logit policy at 0.50. Pairwise significance confirms that this is a real regime split, not just noise:

- KCD vs geometry at 0.35: $\Delta\Delta_{\ell_2} = -53.675$, $p = 0.0272$
- KCD vs geometry at 0.50: $\Delta\Delta_{\ell_2} = +24.357$, $p = 0.0063$

Under scarcity on the hard set, the sparse geometry codec wins. Once budgets loosen, direct geometry retention retakes the lead. The policy reading is therefore explicit but local: `geometry_keep_compress_drop` is the current hard-set low/mid-budget sparse codec, plain `geometry` is the higher-budget fallback/control policy, and `geometry_segment_actions` remains viable but is not

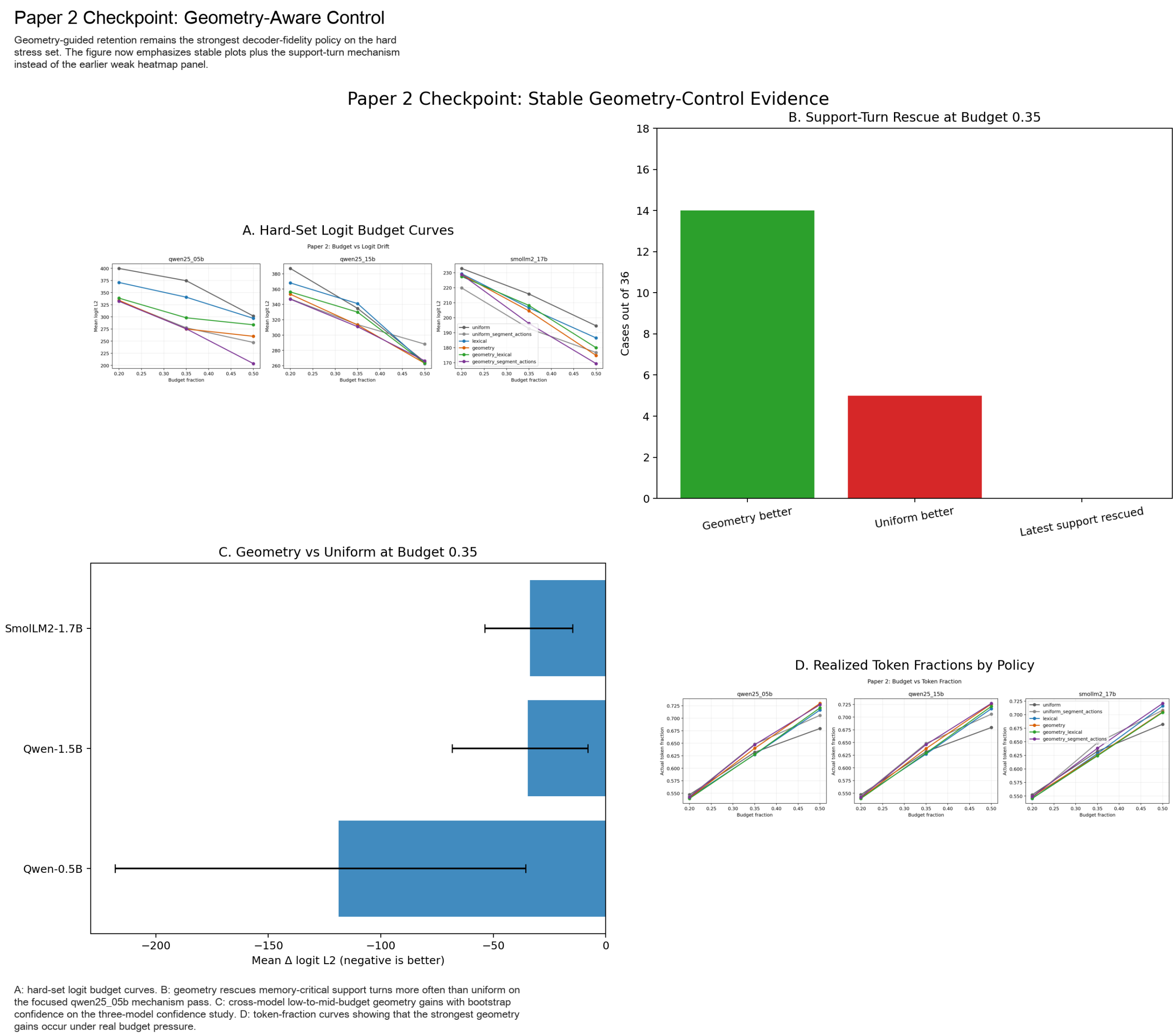

Fig. 2. Paper 2 checkpoint overview. The geometry-family policies consistently outperform uniform in the scarcity regime on the hard stress set. The figure emphasizes stable logit and token-budget curves plus the support-turn rescue mechanism, rather than the earlier weak family-heatmap panel.

the lead family on the fairness-controlled Qwen runs. All comparisons here are budgeted rather than perfectly token-matched; the fairness sweeps are included precisely to test whether the low/mid-budget KCD result survives under tighter realized-token control, and it does.

D. Paper 3 Produces a Benchmark-Dependent Regime Map

The hard-set win does not extrapolate unchanged to broader conversational-memory benchmarks. The public LongMemEval benchmark, MSC-style multi-session conversational memory, and bounded LoCoMo probes all shift the ranking. Table V summarizes the current checkpoint reading.

The clearest public-benchmark result is the full `qwen25_15b` LongMemEval run. At budget 0.20, `semantic` is strongest ($\Delta\Delta_{\ell_2} = -122.613$, $p = 0.0000$). At budget 0.35, `semantic` and `geometry_segment_actions` are strongest (-104.183 and -97.432, both $p = 0.0000$). At budget 0.50, KCD becomes the strongest public-benchmark policy ($\Delta\Delta_{\ell_2} = -48.466$, $p = 0.0000$), while plain geometry becomes significantly worse than uniform ($+31.038$, $p = 0.0032$). This means the geometry program still helps at 0.50 budget, but the winning policy is benchmark-dependent: KCD is the best public-benchmark policy at 0.50, while geometry is the best hard-set 3B policy at the same budget.

MSC pushes the semantic-memory story further. On the 24-conversation `msc_valid` run with `qwen25_15b`, plain `semantic` is the strongest overall family. At budget 0.20, `semantic_keep_compress_drop` is directionally better than plain `semantic` on mean logit distortion, but not significantly so. At 0.35 and 0.50, plain semantic is better overall, especially on behavior. Geometry remains useful relative to

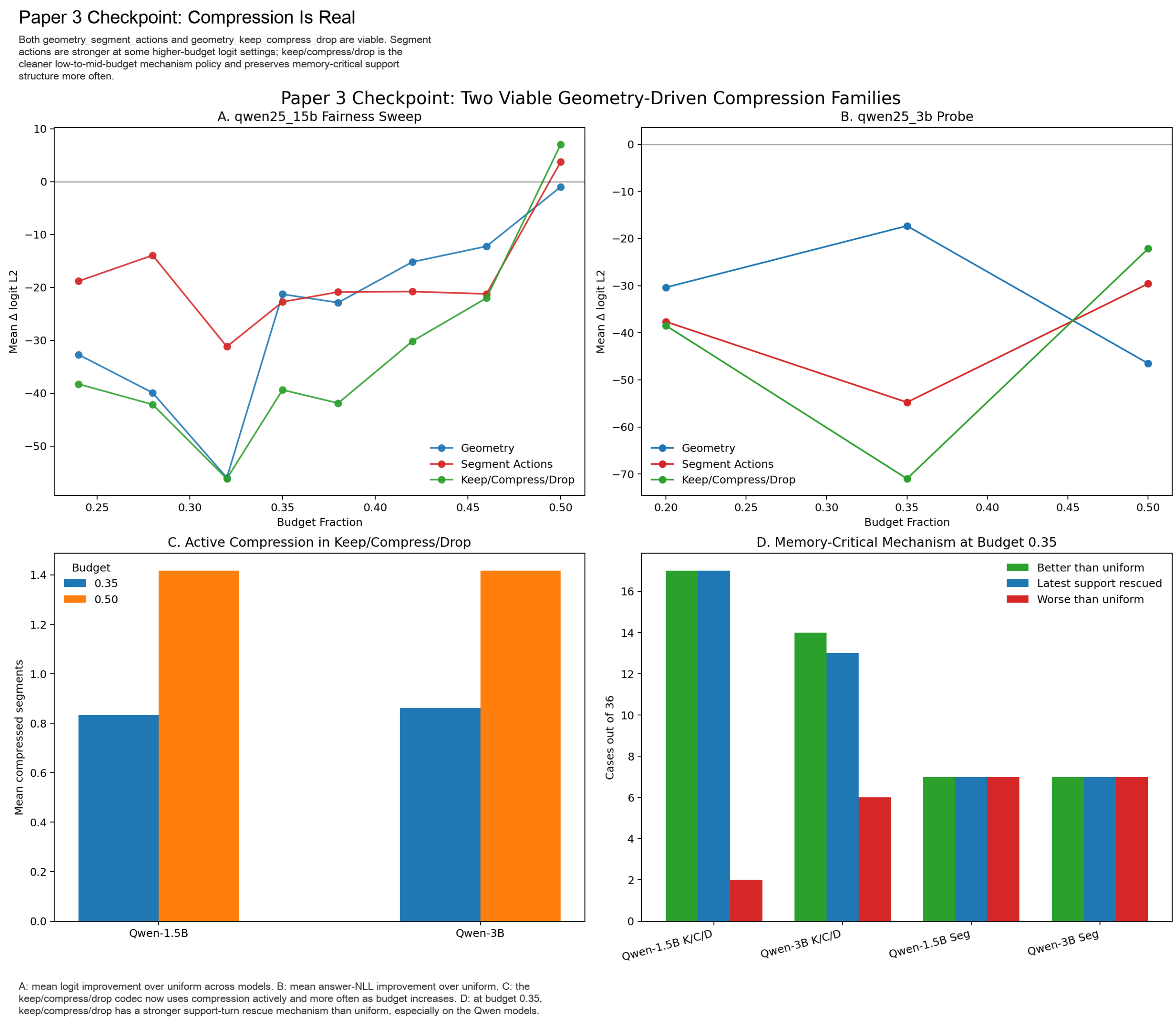

Fig. 3. Paper 3 hard-set checkpoint overview. On the support-turn-critical benchmark family, KCD is the strongest low/mid-budget sparse codec under fairness control, while plain geometry becomes the stronger higher-budget fallback on the 3B probe.

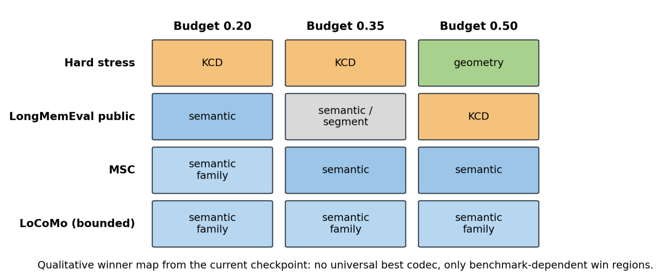

Fig. 4. Qualitative Paper 3 winner map from the current checkpoint. Hard support-turn benchmarks and broader semantic-memory benchmarks do not share the same best policy family.

uniform on MSC, but geometry-KCD is strongly mismatched.

Bounded LoCoMo probes reinforce the same qualitative reading rather than reversing it. The old geometry-KCD family performs poorly, support-aware geometry variants rescue part of that failure, and semantic-led codecs remain strongest overall. Taken together, these results support a benchmark taxonomy rather than a universal winner:

- **support-turn-critical memory** $\rightarrow$ geometry-aware sparse codecs;
- **semantic or persona continuity** $\rightarrow$ semantic-led retention or semantic-led codecs;
- **mixed temporal-semantic memory** $\rightarrow$ still unresolved, but semantic-led families are currently safer than pure geometry codecs.

TABLE V

CURRENT PAPER 3 REGIME MAP. THE WINNING POLICY FAMILY DEPENDS ON THE MEMORY TYPE EMPHASIZED BY THE BENCHMARK.

Evaluation set	0.20	0.35	0.50	Reading
Hard stress / fairness / 3B	KCD	KCD	geometry	Support-turn-critical memory; sparse geometry codecs help most under scarcity.
LongMemEval public	semantic	semantic / segment	KCD	Broader episode memory; geometry-family policies remain useful but ranking changes.
MSC	semantic family	semantic	semantic	Persona and conversational continuity dominate; geometry-KCD is the wrong family.
LoCoMo (bounded)	semantic family	semantic family	semantic family	Mixed temporal-semantic memory; semantic-led codecs remain strongest in current probes.

TABLE VI

NEGATIVE-RESULT CHECKPOINTS. THESE TESTS SHARPEN THE CLAIM BOUNDARY OF THE GEOMETRY PROGRAM RATHER THAN WEAKENING IT.

Check	Setting	Quantitative result	Reading
MSC support vs filler curvature	5 conv.	$\Delta\kappa = 0.122$, 3/5 positive	No robust within-topic curvature separation.
Regime atlas	208 segments	retrieval_heavy: 6 in regime 1, 6 in regime 3	After stabilization, geometry-only regimes still mix stress and casual segments.
State-update alignment	10 synthetic conv.	mean $A = 0.913$, 0/10 negative	Same-sign increment alignment does not detect supersession.
State/increment cross-term	10 synthetic conv.	update 0.971 vs control 0.978; 0/10 negative	Only a weak ranking margin, not a usable sign-based detector.

E. Mechanism: The Codec Preserves the Right Support Structure

Mechanism analysis is the clearest reason not to treat Paper 3 as just another compression heuristic. At budget 0.35:

- on `qwen25_15b`, KCD retains more support user turns than uniform in 17/36 cases, versus 2/36 in the opposite direction;
- on `qwen25_3b`, KCD retains more support user turns than uniform in 14/36 cases, versus 6/36 in the opposite direction;
- compressed cases are not worse than uniform on support retention in 27/29 (`qwen25_15b`) and 23/29 (`qwen25_3b`) of the cases where compression is actually used.

The rescued support objects are semantically precise: code constraints, retrieval packets, exact ordering requirements, and compact fact bundles. That makes the codec interpretable in a way generic token pruning is not. The system is not merely "keeping more useful tokens" in the abstract; it is preferentially preserving the specific earlier turns that later answers depend on.

F. Negative Results and Explicit Limits

The checkpoint is stronger because several plausible extensions failed. Table VI records the main falsification results.

The first failed direction is within-topic curvature discrimination on MSC. In a manual five-conversation support-vs-filler check, mean support-filler curvature delta was only 0.122, with positive deltas in 3/5 conversations and negative deltas in 2/5. That is not a robust support detector.

The second failed direction is geometry-only regime classification. After fixing a genuine long-conversation curvature blow-up, the regime atlas still remained too coarse: retrieval-heavy hard-set segments were split across the same spike-heavy regime family as large portions of MSC and LoCoMo, so geometry alone did not recover a reliable benchmark taxonomy.

The third failed direction is state-update supersession detection. The direct same-sign alignment hypothesis fails completely on a synthetic benchmark of explicit state updates: mean alignment is 0.913 and 0/10 alignments are negative. A more geometrically careful state-vs-increment cross-term also fails as a sign-based detector: mean update cross is 0.971 versus mean control cross 0.978, with 0/10 negative update crosses. The only surviving effect is a very small ranking margin, which is not strong enough to justify a benchmark-scale algorithm.

VI. DISCUSSION

A. What Is Scientifically Settled

The checkpoint is already strong enough to support several stable claims.

1) **Geometry is real and useful.** Paper 1 rules out the strongest null explanations for low-rank conversation dynamics and shows extreme geometry–decoder coupling.
2) **Geometry helps under memory scarcity.** Paper 2 establishes a real geometry-aware control result with a mechanism story.
3) **Compression itself is now contributing.** Paper 3 has an active, fairness-controlled, budget-responsive codec family rather than just a smarter retention policy.
4) **There is no universal best memory signal.** Hard support-turn benchmarks and semantic-memory benchmarks reward different policy families.
5) **Several plausible geometry-only extensions fail in compact models.** Those failures are now part of the scientific result, not side notes.

B. What Remains Open

Three main questions remain open.

1) **Can a learned selector combine the right signals?** The current data suggests semantic candidate discovery plus structure-aware refinement, rather than one fixed codec, but that selector is not learned yet.
2) **What is the model-size threshold for richer geometric discrimination?** The failed state-update and MSC-support checks may be compact-model limitations rather than universal impossibilities.
3) **Can the harm predictor be learned directly?** The theorem directions for Paper 3 still assume bounded distortion terms rather than deriving them from a trained predictor.

C. Limitations

This checkpoint remains a research program in progress, not a finished product.

- The primary mechanism and codec evidence comes from custom hard stress sets designed to expose support-turn failures.
- Most stable results are on compact models, with only the first 3B validations completed.
- Boundary-localization evidence from Paper 1 is secondary and should not be oversold as exact regime recovery.
- Several broader-benchmark codec results are still bounded checkpoint studies rather than fully standardized large-scale sweeps.
- The later negative-result checks were designed to falsify plausible extensions quickly; they sharpen the claim boundary, but do not by themselves provide an alternative algorithm.

VII. Reproducibility and Repository Packaging

This manuscript is paired with a versioned repository checkpoint that includes:

- frozen artifact bundles for Papers 1–3 under `artifacts/`,
- Colab notebooks for local and cloud reruns under `notebooks/`,
- runner scripts under `scripts/`,
- manuscript assets generated directly from tracked JSON summaries under `manuscript/generated/`.

The manuscript tables in this PDF are generated from tracked artifact files, not manually transcribed. The build entry point is:

```
python scripts/build_manuscript_assets.py
tectonic manuscript/paper_checkpoint.tex
```

VIII. Conclusion

The current checkpoint shows that geometry-aware conversational memory compression is a coherent, cumulative program rather than a disconnected set of experiments, but the final claim is now more precise than the early optimistic version. Paper 1 establishes a strong geometric substrate: conversation trajectories are extremely low-rank and almost perfectly decoder-coupled. Paper 2 turns that substrate into a real control law under scarcity, with a concrete mechanism centered on support-turn rescue. Paper 3 shows that sparse codecs are real and useful on support-turn-critical benchmarks, and that the geometry program still helps at 0.50 budget, but it also shows that there is no universal best policy family. Semantic-memory benchmarks favor semantic-led methods; several tempting geometry-only extensions fail. That is a stronger research outcome than a single unchecked win story. The right next step is not to keep searching for another standalone geometry heuristic. It is to build a benchmark-aware or learned selector that uses geometry where structure matters, uses semantic signal where broad continuity matters, and treats the negative results in this checkpoint as hard design constraints rather than inconveniences.

Appendix A
Storage Accounting for Sparse Codecs

Any honest codec comparison must count more than just retained raw tokens. For segment j, the storage model should include the anchor state, sparse support memory, and metadata:

$$\text{Storage} = \sum_j \left(\underbrace{\|a_j\|}_{\text{anchor}} + \underbrace{\|s_j\|}_{\text{sparse support}} + \underbrace{\|\text{meta}_j\|}_{\text{segment metadata}} \right). \quad (14)$$

Future work should refine this accounting into exact byte counts for latent summaries and decoder-compatible surrogates.

Appendix B
Checkpoint Artifact Map

The most important tracked bundles are:

- Paper 1 final: `artifacts/paper1/expanded_v8_final`
- Paper 1 regime atlas limits: `artifacts/paper1/regime_atlas_smoke_v4`
- Paper 2 hard stress: `artifacts/paper2/behavior_stress_v1`
- Paper 2 mechanism cases: `artifacts/paper2/behavior_stress_qwen_cases`
- Paper 3 fairness: `artifacts/paper3/paper3_batch_v1_fairness`
- Paper 3 3B probe: `artifacts/paper3/paper3_batch_v1_3b`
- Paper 3 MSC persona falsification: `artifacts/paper3/msc_persona_curvature_v1`
- Paper 3 state-update falsification: `artifacts/paper3/state_update_alignment_smoke_qwen05b`
- Paper 3 state/update cross control: `artifacts/paper3/state_update_cross_control_qwen05b`

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